



12.3.7 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson?



Unit 12, Lesson 4: Displaying and Interpreting Univariate Categorical Data



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



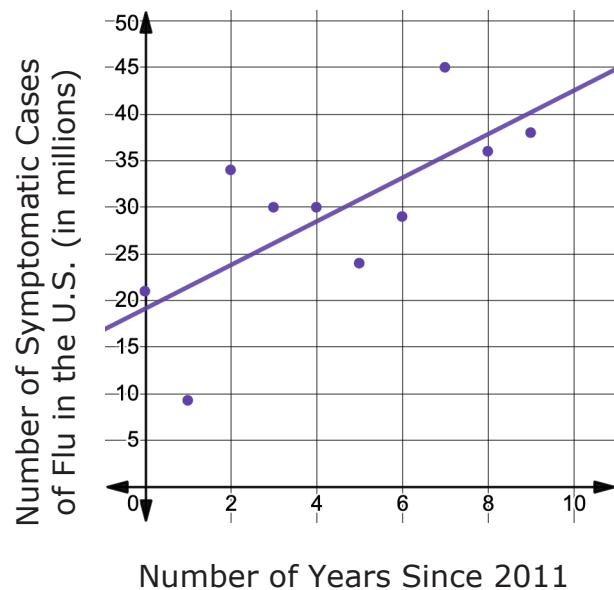
Learning Targets

- ☐ I can create an appropriate graphical display for univariate categorical data.
- ☐ I can interpret a graphical display for univariate categorical data.



12.4.1 Warm-Up: Line of Fit on a Scatter Plot

A scatter plot of the data on symptomatic cases of the flu in the U.S. is shown. The line of fit, $y = 2.34x + 19.1$, is included on the scatter plot, where y is the number of symptomatic cases of the flu in the U.S., in millions, and x is the number of years since 2011.



1. Interpret the slope of the line of fit.

2. Interpret the y -intercept of the line of fit.



12.4.2 Guided Instruction: Displays for Categorical Data

A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology.

1. The results for 741 teens who use social media, when asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral), is summarized.

- 219 said the effect was mostly positive
- 211 said the effect was mostly negative
- 311 said the effect was neutral

a. Which of the following display(s) can be used to represent the data? Select all that apply.

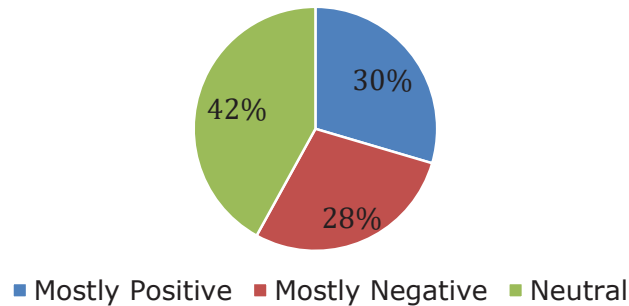
- ☐ Box plot
- ☐ Histogram
- ☐ Circle graph
- ☐ Frequency table
- ☐ Relative frequency table
- ☐ Bar graph
- ☐ Scatter plot

b. Create a bar graph to display the data.



2. The same data is summarized in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



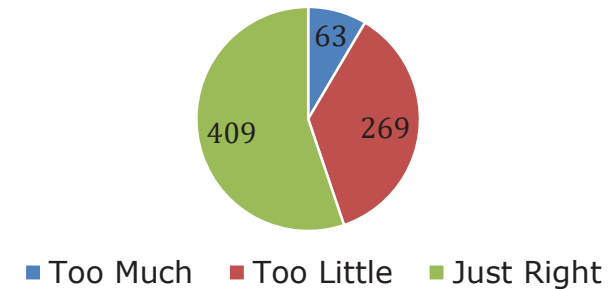
The percentage of teens who describe social media as having a mostly positive effect on people

their age is ☐ less than ☐ equal to ☐ greater than the percent of students who believe it has had a negative effect.

On the same survey, data was also collected on how much time they believe they spend with their friends in person (too much, too little, or about the right amount).

3. The data is summarized in the circle graph.

Time Spent With Friends in Person



What percentage of participants feel that they spend too little time with their friends in person?

4. What do you know about the relationship between time spent with friends in person and perceived effect of social media?

5. To analyze the relationship between how much time they spend with their friends and their perceived effect social media has on people their age, an appropriate

display for ☐ numerical, ☐ categorical, ☐ univariate, ☐ bivariate

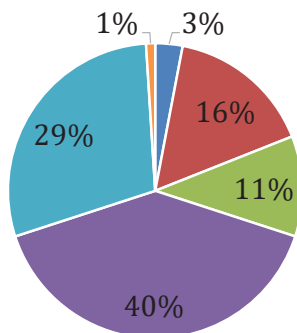
data should be created.



12.4.3 Your Turn!

1. In the 2018 Pew Research Center survey of American teenagers, 634 respondents used more than one social media site. Data on their most used site is summarized in the circle graph.

Most Used Social Media Site



■ Twitter ■ Instagram ■ Facebook
■ Snapchat ■ YouTube ■ Reddit

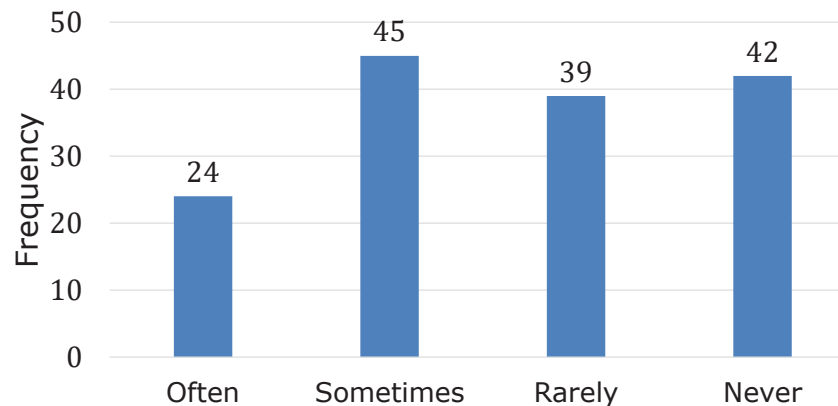
- a. What was the most used social media site?
- b. What was the least used social media site?
- c. How many respondents said Instagram was their most used site?



12.4.4 Wrap-Up: Error Analysis

1. A survey of U.S. teens asked participants how often they post selfies. The results are summarized in the bar graph.

How Often Teens Post Selfies



Rayah analyzed the bar graph and concluded that the majority of U.S. teens in the sample post selfies sometimes or often since the percent who post often plus the percent who post sometimes is more than 50%.

- a. Explain the error in Rayah's interpretation.
- b. Find the percent of U.S. teens in the sample who post selfies sometimes or never.